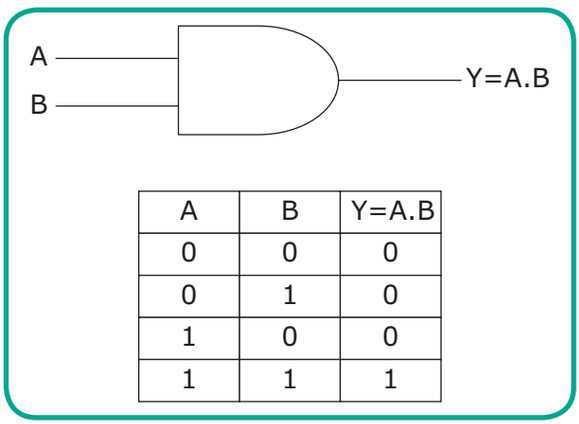
The two input signal makes 2^2 combination of outputs.

In OR gates, the output is high when any one of the input is in high level. Conversely, the output is low when all the inputs are in low level.

8.7.3 AND Gate

An AND gate has two or more inputs and one output. An AND gate performs logical multiplication. In an AND gates, the inputs A, B, C, etc., produce the output as A.B.C.etc. The symbol and the truth table of two input AND gate are shown in Figure 8.13.

It contains two input signals and only one output signal. In AND gates, the output is only high when all inputs are in high level. Conversely, the output is low only when any one of the input is in low level.



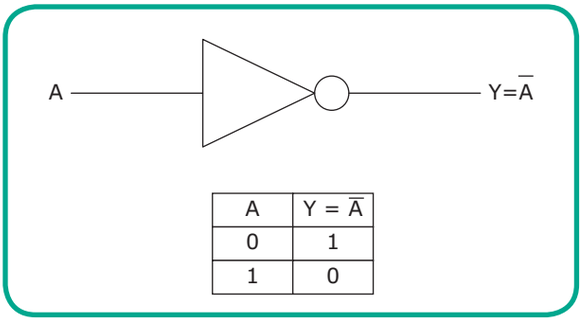
A	B	Y=A.B
0	0	0
0	1	0
1	0	0
1	1	1

Figure 8.13 Symbol and Truth Table of AND Gate

8.7.4 NOT Gate

A NOT gate has only one input and one output. For the NOT gate, when the input is '0' (LOW), the output is '1' (HIGH)and when the input is '1' (HIGH), the output is '0' (LOW). That is, the output is complement or inverse of the input.

Figure 8.14 shows the symbol and truth table for the NOT gate. The input is marked as A and the output is marked as $Y=\overline{A}$. The output can be read as complement of A or inverse of A or simply A bar. It is also known as inverter.



A	Y = $\overline{A}$
0	1
1	0

Figure 8.14 Symbol and Truth Table of NOT Gate

8.8 BOOLEAN ALGEBRA

Digital circuits perform the binary arithmetic operations with binary digits 1 and 0. These operations are called logic functions or logic operations. *The algebra used to symbolically describe logic function is called* **Boolean algebra.**

Boolean algebra is a set of rules and theorems by which logical operations can be expressed symbolically in equation form and be manipulated mathematically. As with the ordinary algebra, the letters of alphabet (e.g. A, B, C, etc.) can be used to represent the variables. Boolean algebra differs from ordinary algebra, since the Boolean constant and variables can have only two values 0 and 1. There are four connecting symbols used in Boolean algebra viz.

i) equals sign(=) ii) plus sign(+)

iii) Multiply sign (.) iv) bar (-)

1. Equals Sign (=): The equals sign in Boolean algebra refers to the standard mathematical equality. In other words, the logical value on one side of the sign is identical to the logical value on the other side

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